Exam 1 solutions

① $(x\ln x)y' + y = 2\ln x, \quad y(e) = 2$

First, divide both sides by $x\ln x$

$$y' + \frac{1}{x\ln x}\, y = \frac{2}{x}$$

$$u = e^{\int \frac{dx}{x\ln x}} = e^{\ln \ln x} = \ln x$$

(we are told that $x>0$ and $\ln x > 0$ so we don't need to use the abs. value signs)

Multiply both sides by $\ln x$

$$y' \ln x + \frac{1}{x} y = \frac{2\ln x}{x}$$

$(y\ln x)' = \frac{2\ln x}{x}$. Integrate both sides

$$y\ln x = (\ln x)^2 + C \Rightarrow y = \ln x + \frac{C}{\ln x}$$

$y(e) = 2 \Rightarrow 2 = \ln e + \frac{C}{\ln e}$, or $2 = 1 + C \Rightarrow C = 1$

$$y = \ln x + \frac{1}{\ln x}$$

② $\underbrace{\left(\frac{y}{x} + 6x\right)}_{M} dx + \underbrace{(\ln x - 2)}_{N} dy = 0$

We check if this equation is exact:

$\frac{\partial M}{\partial y} = \frac{1}{x}$ and $\frac{\partial N}{\partial x} = \frac{1}{x}$. Yes, this equation is exact!

We need to find F such that

$\frac{\partial F}{\partial x} = \frac{y}{x} + 6x$ and $\frac{\partial F}{\partial y} = \ln x - 2$

We integrate the first equation (with respect to x)

$F = y \ln x + 3x^2 + g(y)$. Now we calculate

$\frac{\partial F}{\partial y}$ and compare it to N:

$\frac{\partial F}{\partial y} = \ln x + g'(y) = \ln x - 2 \Rightarrow g'(y) = -2 \Rightarrow$

$g(y) = -2y + C \Rightarrow F = y \ln x + 3x^2 - 2y + C.$

The general solution is

$y \ln x + 3x^2 - 2y + C = C_1$, or

$y \ln x + 3x^2 - 2y = C_2$

(where $C_2 = C_1 - C$)

(3) a) Let A = amount of chemical. We know that

$$\frac{dA}{dt} = \text{rate in} - \text{rate out}$$

$$\text{rate in} = 2\,\frac{lb}{gal} \cdot 2\,\frac{gal}{min} = 4\,\frac{lb}{min}$$

to calculate rate out, we first calculate the volume. Each minute, we have 2 gallons entering the tank and 3 gallons leaving the tank, so we are losing 1 gallon each minute. After t minutes, we've lost t gallons, so volume $= 100 - t$.

$$\text{rate out} = \frac{A}{100-t}\,\frac{lb}{gal} \cdot 3\,\frac{gal}{min} = \frac{3A}{100-t}\,\frac{lb}{min}$$

$$\frac{dA}{dt} = 4 - \frac{3A}{100-t}, \quad A(0) = 40$$

$$\frac{dA}{dt} + \frac{3}{100-t}A = 4$$ This is a linear equation

$$u = e^{\int \frac{3}{100-t}dt} = e^{-3\ln(100-t)} = e^{\ln(100-t)^{-3}}$$

$$= (100-t)^{-3} = \frac{1}{(100-t)^3}.$$

Multiply both sides by $\frac{1}{(100-t)^3}$

$$\frac{1}{(100-t)^3} \cdot \frac{dA}{dt} + \frac{3}{(100-t)^4}A = \frac{4}{(100-t)^3}$$

$$\left(\frac{1}{(100-t)^3}A\right)' = \frac{4}{(100-t)^3}$$ Integrate both sides

$$\frac{1}{(100-t)^3}A = \frac{2}{(100-t)^2} + C \Rightarrow$$

$$A = 2(100-t) + C(100-t)^3.$$

We know $A(0) = 40 \Rightarrow$

$$40 = 200 + C \cdot 100^3 \Rightarrow C = -\frac{160}{100^3}$$

$$\Rightarrow A = 2(100-t) - \frac{160}{100^3}(100-t)^3$$

So we get $A(50) = 2(50) - \frac{160}{100^3} \cdot 50^3$

$$= 100 - \frac{160}{8} = 100 - 20 = 80 \text{ lbs}$$

③ b) We use the Newton's law of cooling

$$\frac{d\theta}{dt} = k(\theta - 75).$$ We are given $\theta(0) = 150$ and $\theta(10) = 125$.

$$\frac{d\theta}{\theta - 75} = k\,dt.$$ Integrate both sides

$$\ln|\theta - 75| = kt + C \Rightarrow |\theta - 75| = e^{kt + C}$$

$$\Rightarrow \theta - 75 = \pm e^{C} \cdot e^{kt} = C_1 e^{kt} \quad (\text{where } C_1 = \pm e^{C})$$

$$\theta = 75 + C_1 e^{kt}$$

$$\theta(0) = 75 + C_1 = 150 \Rightarrow C_1 = 75 \Rightarrow$$

$$\theta = 75 + 75 e^{kt}$$

$$\theta(10) = 125 = 75 + 75 e^{10k} \Rightarrow 50 = 75 e^{10k}$$

$$e^{10k} = \frac{50}{75} = \frac{2}{3} \Rightarrow 10k = \ln\frac{2}{3} \Rightarrow k = \frac{1}{10}\ln\frac{2}{3}$$

$$\theta = 75 + 75 e^{\frac{1}{10}\ln\frac{2}{3}\cdot t}.$$ We need to find t such that $\theta = 100$

$$75 + 75 e^{\frac{1}{10}\ln\frac{2}{3}t} = 100$$

$$75 e^{\frac{1}{10}\ln\frac{2}{3}t} = 25 \Rightarrow$$

$$e^{\frac{1}{10}\ln\frac{2}{3}\cdot t} = \frac{25}{75} = \frac{1}{3} \Rightarrow$$

$$\frac{1}{10}\ln\frac{2}{3}\cdot t = \ln\frac{1}{3} \Rightarrow$$

$$t = \frac{10\ln\frac{1}{3}}{\ln\frac{2}{3}} \approx 27.1 \text{ min}$$

4 a) $y'' = y' + 3$

Let $v = y'$; $v' = y''$. We get

$v' = v + 3. \Rightarrow v' - v = 3.$ This is a linear equation.

$u = e^{\int -1\,dx} = e^{-x}$. Multiply both sides by e^{-x}

$v'e^{-x} - ve^{-x} = 3e^{-x}$, or

$(ve^{-x})' = 3e^{-x}$. Integrate both sides

$ve^{-x} = -3e^{-x} + C \Rightarrow v = -3 + Ce^{x}$

This means $y' = -3 + Ce^{x}$. Integrate this to get $y = -3x + Ce^{x} + C_1$.

4b) $y' + \frac{3}{x} y = x^2 y^2$

This is a Bernoulli equation

We divide both sides by y^2

$\frac{y'}{y^2} + \frac{3}{x} \cdot \frac{1}{y} = x^2$. Let $v = \frac{1}{y}$. Then $v' = -\frac{y'}{y^2}$

$\Rightarrow \frac{y'}{y^2} = -v'$

We get $-v' + \frac{3}{x} v = x^2$, or

$v' - \frac{3}{x} v = -x^2$. This is a linear equation

$u = e^{\int -\frac{3}{x} dx} = e^{-3 \ln x} = e^{\ln x^{-3}} = x^{-3} = \frac{1}{x^3}$

Multiply both sides by $\frac{1}{x^3}$ $\Rightarrow$

$v' \cdot \frac{1}{x^3} - \frac{3}{x^4} v = -\frac{1}{x}$, or $\left(v \cdot \frac{1}{x^3}\right)' = -\frac{1}{x}$.

Integrate both sides:

$v \cdot \frac{1}{x^3} = -\ln x + C \Rightarrow v = (C - \ln x) x^3$

Now, $v = \frac{1}{y} \Rightarrow y = \frac{1}{v} = \frac{1}{(C - \ln x) x^3}$

(5) $y' = y + 1 - x, \quad y(0) = 0$

$y_0 = 0 \qquad f(x, y) = y + 1 - x$

$$y_1 = 0 + \int_0^x f(t, 0)\,dt = \int_0^x (1 - t)\,dt =$$

$$= \left[t - \frac{t^2}{2}\right]_0^x = x - \frac{x^2}{2}$$

$$y_2 = 0 + \int_0^x f\left(t, t - \frac{t^2}{2}\right)dt = \int_0^x \left(t - \frac{t^2}{2} + 1 - t\right)dt$$

$$= \int_0^x \left(1 - \frac{t^2}{2}\right)dt = \left[t - \frac{t^3}{6}\right]_0^x = x - \frac{x^3}{6}$$

$$y_3 = 0 + \int_0^x f\left(t, t - \frac{t^3}{6}\right)dt = \int_0^x \left(t - \frac{t^3}{6} + 1 - t\right)dt$$

$$= \int_0^x \left(1 - \frac{t^3}{6}\right)dt = \left[t - \frac{t^4}{24}\right]_0^x = x - \frac{x^4}{24}$$